

BEYOND NODAL DAMPING

WHAT LOCAL PLL RETUNING CAN AND CANNOT DO

Exact delay compensation and network-mediated modal spillover
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1 INTRO AND OBJECTIVES

Introduction

Inverters (GFL) lock to the grid with a phase-locked loop (PLL). Measurement and processing add latency. A few extra milliseconds can make a converter-dominated grid oscillate.

The usual answer is to retune the PLL gains k_p, k_I so that a chosen oscillation mode is unchanged.

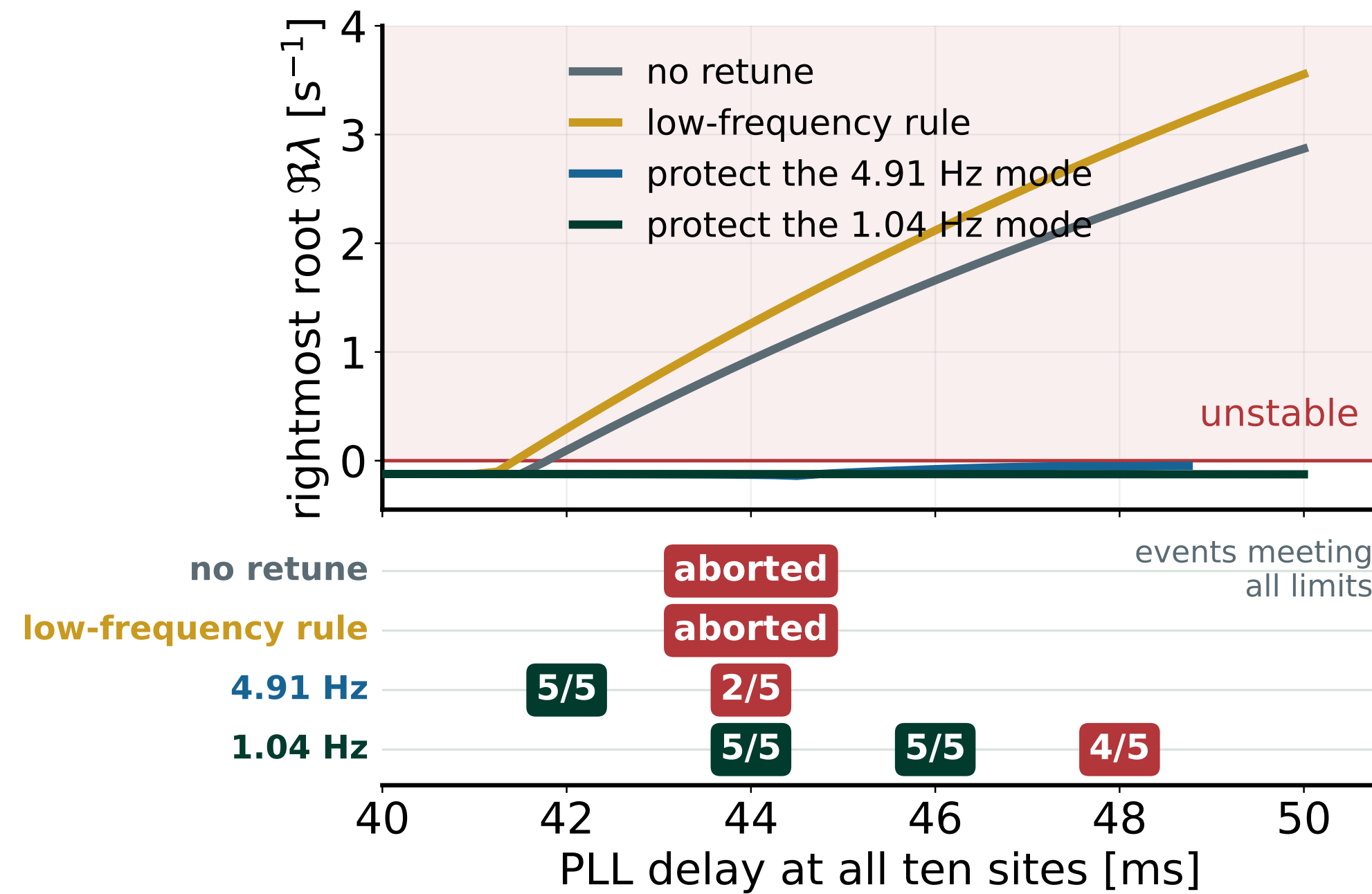
Objectives

- 1 Find the exact gains that hide a delay change from one mode.
- 2 Compute what that retune does to every other mode.
- 3 Use it to choose which mode to protect, and check spectrum and events.

IN WORDS If the retune keeps one mode exactly where it was, is the rest of the grid safe?

Setting. IEEE-39, ten SG/GFL sites at 87.5% inverter share, PLL delay 40 ms raised at all sites, five nonlinear events.

2 WHAT WE FOUND: WHICH MODE YOU PROTECT DECIDES



At 44 ms (was 40 ms)

rule	roots	events
no retune	8 / 10	aborted
low-frequency	12 / 12	aborted
protect 4.91 Hz	0 / 0	2 of 5
protect 1.04 Hz	0 / 0	5 of 5

roots: unstable / beyond the -0.05 s^{-1} margin, full-spectrum count. Events: five nonlinear load steps with frequency, RoCoF, voltage and SG-reserve limits.

IN WORDS The usual low-frequency compensation is worse than no retune. Protecting the least-damped 4.91 Hz mode restores the spectrum but uses up the frequency-response reserve. Protecting a well-damped 1.04 Hz mode keeps both.

3 THE LAW

PLL loop factor: $\kappa(s) = (k_p s + k_I) e^{-s\tau}$. Let the delay grow by h and pick a mode $\lambda = \alpha + j\omega$ to protect.

Exact transport of the gains

$$k_p(h)\lambda + k_I(h) = (k_p\lambda + k_I)e^{\lambda h}$$

What is left at every other frequency

$$\Delta\kappa(s) = -(s - \lambda)(s - \bar{\lambda})e^{-s\tau} \int_0^h k_p(t)e^{-st} dt$$

IN WORDS Two real gains match the old loop at one complex frequency. The error is zero at the protected mode and grows quadratically away from it.

Unique if $\omega \neq 0$. All sites using the same λ keep it exactly, for any delays h_i . Derivation: differentiate κ ; the gain flow has eigenvalues $\lambda, \bar{\lambda}$.

1 DELAY GROWS

$$\tau \rightarrow \tau + h$$



2 TRANSPORT

$$k\lambda \rightarrow k\lambda e^{\lambda h}$$



3 SPILLOVER

$$(\mu - \lambda)(\mu - \bar{\lambda}) \partial_{k_I} \mu$$



4 CHOOSE AND CHECK

spectrum and events

4 HOW: THE GRID SETS THE PRICE, AND THE PRICE CHOOSES THE MODE

A. Spillover onto any other mode μ

$$\frac{d\mu}{dh} = -k_p(\mu - \lambda)(\mu - \bar{\lambda}) \frac{\partial \mu}{\partial k_I}$$

The PLL fixes the quadratic factor. The grid fixes the residue, through its reduced operator $Z(s)$ with left and right vectors ℓ, r :

$$\frac{\partial \mu}{\partial k_I} = -\frac{\ell^H Z_{k_I}(\mu) r}{\ell^H Z_s(\mu) r}$$

IN WORDS Preserving a mode is exact and local. Whether it is safe is decided elsewhere: by how strongly the network couples the retuned PLLs to every other mode, and by how much integral gain the choice spends.

5 RESULTS: SPECTRUM AND NONLINEAR EVENTS, IEEE-39

rule	ms	k_I	roots	events	$ \Delta f $ [Hz]	SG reserve
no retune	42	246.7	4 / 4	1 of 1 [†]	0.444	0.0074
no retune	44	246.7	8 / 10	aborted	—	—
low-frequency	44	246.7	12 / 12	aborted	—	—
protect 4.91 Hz	42	192.5	0 / 0	5 of 5	0.486	0.0032
protect 4.91 Hz	44	137.8	0 / 0	2 of 5	0.505	≈ 0
protect 1.04 Hz	44	213.3	0 / 0	5 of 5	0.480	0.0047
protect 1.04 Hz	46	197.8	0 / 0	5 of 5	0.485	0.0033
protect 1.04 Hz	48	183.0	0 / 0	4 of 5	0.491	0.0018

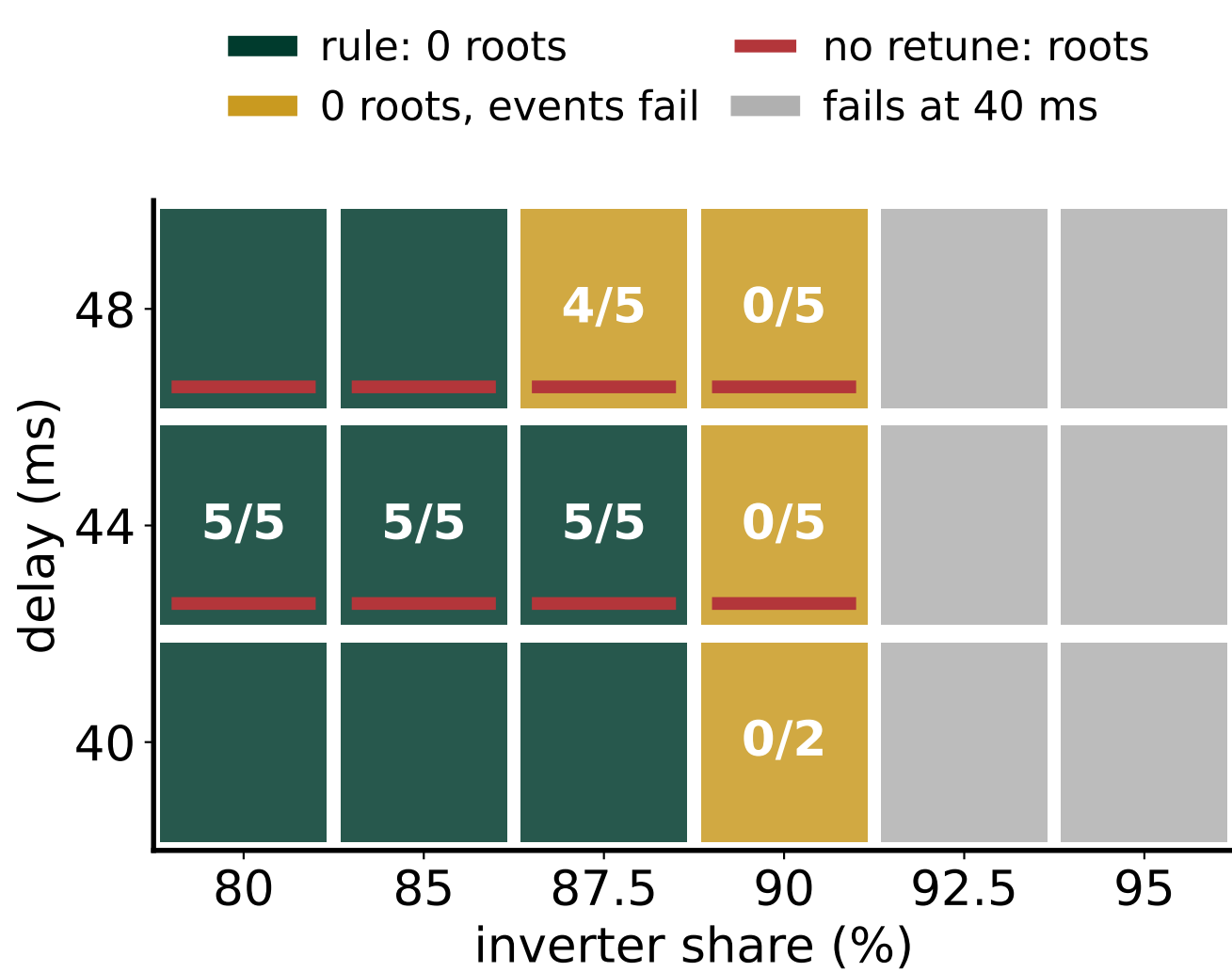
Limits: 0.5 Hz, 0.5 Hz/s, $V \in [0.9, 1.1]$, SG reserve ≥ 0.002 . Worst value over the events run. Delayed nonlinear model, method of steps, 60 s. “Aborted”: the integrator stopped on the events tried. [†]One 30 s event meets the limits although four roots are unstable ($+0.096 \text{ s}^{-1}$ at 4.8 Hz): 0.5 s windows do not see slow 5 Hz growth, so events are always read together with the root count.

7 HOW FAR IT REACHES: INVERTER SHARE $\times$ DELAY

Each cell retunes the PLLs by the rule and counts roots beyond the margin. Red bars mark the same cell without retuning: roots appear at 44 and 48 ms. Numbers are nonlinear events passed.

IN WORDS The rule extends latency tolerance at every share where the 40 ms design is feasible; it does not raise the maximum share (92.5% already fails at 40 ms).

At 90% the 40 ms design with its original gains already fails the events (0 of 2, up to 0.58 Hz). The retuned 44 and 48 ms cases fail the same way: the limit there is set by the inverter share, not by the retune.



The damping-signature result (δD_H has one positive and one negative eigenvalue) is in the companion paper.

6 DOES THE RULE TRANSFER?

Second base design (85% inverter share at every site). The selection rule was fixed on the first design and applied without change; it picks the 0.93 Hz mode.

rule, 44 ms	roots	events	SG reserve
no retune	10 / 12	aborted	—
low-frequency	12 / 12	not run	—
rule-selected mode	0 / 0	5 of 5	0.027

Is the exact formula needed? Its first-order version gives the same root counts at 42–48 ms for both protected modes. At these delays the choice of mode matters; the truncation does not.

Checks of the identities. Spillover formula against finite differences: 219 mode pairs, largest relative error 3.9×10^{-5} . Predicted against exact worst root over 11 candidate modes: rank correlation 0.99.

Third model, re-run here: a nine-bus SG–GFL bank with assumed dynamic data. From 20 to 21 ms the rightmost root is $+0.886 \text{ s}^{-1}$ unchanged and -0.131 s^{-1} with transport on its three PLLs (spectral re-run); the 12 of 12 events are reported, not re-run.

8 LIMITS, AND WHAT DID NOT WORK

Where the rule runs out

Phase budget. Protecting the 4.91 Hz mode drives k_I to zero at $h^* = 8.99$ ms. The spectrum is still inside the margin there, but the events already fail at 44 ms. The budget is a limit of the formula, not an operating limit.

Frequency-response reserve. Every protected mode lowers k_I . The 1.04 Hz choice passes all events up to 46 ms and loses one at 48 ms (SG reserve 0.0018).

What failed

Linear programme over the spillover formula. Extrapolated far from the base point it proposed a design with 16 unstable roots. The formula ranks nearby choices; it does not bound large steps.

One assigned pole is not stability. In the nine-bus bank an exactly assigned pole sits at -0.30 s^{-1} while another pair stays at $+0.57 \text{ s}^{-1}$.

IN WORDS Not claimed: an optimal mode, a maximum tolerable delay, a maximum inverter share, or superiority over other compensators. Root counts are floating-point, checked against eigenvalues at zero delay, not interval-certified.

9 CONCLUSIONS

Exact and local

The gains that hide a delay change from one mode are closed-form and need only the device's own gains and delay.

Priced by the network

Every other mode moves by a known quadratic factor times a collective residue. That price selects the mode to protect.

Checked twice

Spectrum and nonlinear events disagree at 44 ms for one choice. A design rule has to pass both; the low-frequency rule passes neither.

TAKEAWAY

PRESERVING ONE MODE DOES NOT PROTECT THE GRID.

The gains that hide a delay from one mode are exact. The network decides what they do to all the others.

REFERENCES

- [1] Marković et al., IEEE Trans. Power Syst., 2021.
- [2] Huang et al., arXiv:1903.05489.
- [3] Michiels & Gumussoy, arXiv:2003.05496.
- [4] Dörfler et al., IEEE Trans. Power Syst., 2014.
- [5] Bindel & Hood, SIAM J. Matrix Anal. Appl., 2013.
- [6] Johansson, IEEE Trans. Comput., 2017.

Topics: [1] converter small-signal stability; [2] PLL stability and grid structure; [3] fixed-order design with delays; [4] sparse wide-area control; [5] nonlinear eigenvalues; [6] interval arithmetic.

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